

Multiple Choice

1. **Answer:** D

Options: A) $mc/2$; B) $mc/(2^{1/2})$; C) mc ; D) $(3^{1/2})mc$

Note: Correct option: D - $(3^{1/2})mc$

2. **Answer:** A

Options: A) Planck's constant; B) Boltzmann's constant; C) The Rydberg constant; D) The speed of light

Note: Correct option: A - Planck's constant

3. **Answer:** D

Options: A) independent of M ; B) proportional to $m^{1/2}$; C) linear in R ; D) proportional to $R^{3/2}$

Note: Correct option: D - proportional to $R^{3/2}$

4. **Answer:** A

Options: A) 4; B) 5; C) 6; D) 20

Note: Correct option: A - 4

5. **Answer:** D

Options: A) 0.25 W; B) 0.5 W; C) 1 W; D) 4 W

Note: Correct option: D - 4 W

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The entropy of the system and its environment remains unchanged.'

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'gamma rays'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' 1.5×10^8 m/s'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** decreased by a factor of 9. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 10/11 c. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key